

# Internet Landscape Study of Southeast Asian Countries

Chissanapun  
Athiwarikanon  
chissanapun.a@ku.th  
Kasetsart University  
Department of Computer  
Science  
Bangkok, Thailand

Kunanont Malayanont  
kunanont.m@ku.th  
Kasetsart University  
Department of Computer  
Science  
Bangkok, Thailand

Pakkapon Pattanakul  
pakkapon.p@ku.th  
Kasetsart University  
Department of Computer  
Science  
Bangkok, Thailand

Sukumal Kitisin  
sukumal.i@ku.th  
Kasetsart University  
Department of Computer  
Science  
Bangkok, Thailand

## ABSTRACT

National Internet broadband service providers particularly in Thailand are often perceived from user experiences. In Southeast Asian this assumption remains largely unmeasured or studied. For example, Thailand ranked 12th worldwide for fixed broadband in Ookla’s Speedtest Global Index (June 2026) [3], yet whether such the services reflect application-level performance across the country is unclear. A 2023 survey by Thailand Consumer Council found that 81% of 2,924 respondents had encountered connection problems, mostly slow and dropping connections [5]. These numbers contradict with each other for Thailand, and definitely, Thailand is not alone on the mismatched information in Southeast Asian region.

In this study we analyze and evaluate the communication network quality of nine countries: Thailand, Vietnam, Laos, Myanmar, Cambodia, Indonesia, the Philippines, Malaysia, and Singapore using the test results from Q1 of 2023 to Q4 of 2025 using collected open datasets from Ookla (229.4 million tests) and M-Lab NDT7 (760.6 million tests). We compared the results to the bandwidth requirements of four types of popular application namely voice, video streaming (HD), video streaming (UHD), and Cloud gaming.

## KEYWORDS

Speed tests analysis, Ookla, NDT7

## 1 INTRODUCTION

National Internet Service Providers (ISPs), particularly in Southeast Asian countries like Thailand, are frequently judged based on subjective end-user perceptions rather than systematic empirical data. A significant gap exists between high-level speed index rankings and actual user experiences; for example, while Thailand was ranked 12th worldwide for fixed broadband in Ookla’s Speedtest Global Index in June 2026, a 2023 survey by the Thailand Consumer Council revealed that 81% of 2,924 respondents suffered from frequent connection problems such as slow speeds and dropped signals. To address this widespread mismatch across the region, this paper presents a comprehensive study analyzing communication network quality across nine Southeast Asian nations: Thailand, Vietnam, Laos, Myanmar, Cambodia, Indonesia, the Philippines, Malaysia, and Singapore. Leveraging cross-platform open measurement datasets collected from Q1 2023 to Q4 2025—comprising 229.4 million tests from Ookla and 760.6 million tests from Measurement Lab’s NDT7—we evaluate performance against real-world application demands.

**Table 1: Bandwidth and latency requirements for different types of applications**

| Application Types     | Required Data Rate | Required Latency |
|-----------------------|--------------------|------------------|
| Voice                 | 64 Kbps            | 200ms            |
| Video Streaming (HD)  | 5Mbps              | few secs         |
| Video Streaming (UHD) | 25Mbps             | few secs         |
| Cloud gaming          | 44Mbps             | 25 ms            |

## 2 DATA AND METHOD

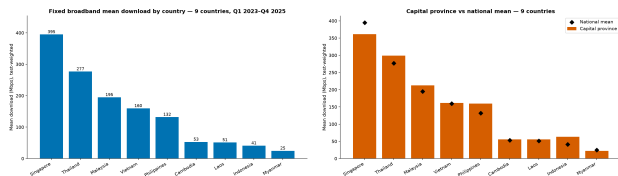
Ookla open data supplies quarterly internet-performance tiles split by connection type (fixed, mobile) at Quadkey zoom 16 (~600x600 m), each carrying mean download, upload, latency, and the underlying Speedtest submission count, delivered as global Parquet files on Amazon S3. We map them to the nine countries and twelve quarters (Q1 2023–Q4 2025); having the total of 229,367,221 tests, with largest Indonesia test data of 74.2 million records and smallest Laos dataset of 875,799 records. Total volume across the nine countries from Mlab NDT7 is 760,607,387 tests, with largest Indonesia test data of 367.2 million and smallest Laos test record of 139,693 records.

The data from the two sources comes in different formats. Ookla comes as spatial tiles approximately 600m x 600m, while NDT7 comes as individual test records. We keep the two sources separate throughout and never pool their records; we analyze each independently and compare the results side by side. First we clean the data such as throughput  $\leq 0$  and duration  $\leq 0$ . We then use ip-api.com [1] which is a free API service for checking whether the test records are the mobile / hosting / proxy flags and the ASN for each IP. On Ookla dataset, we build a bounding box to define each country’s extent, referencing the geoBoundaries ADM1 file [4] for the 9 ASEAN countries, then run point-in-polygon on the tiles to determine which province each point falls in. Then compute a weighted average by each tile’s test count, to prevent distortion from tiles with widely different test counts. To ensure quality, we filter only province-quarters which has total\_tests  $\geq 100$  and n\_tiles  $\geq 5$  which would better represent the reality.

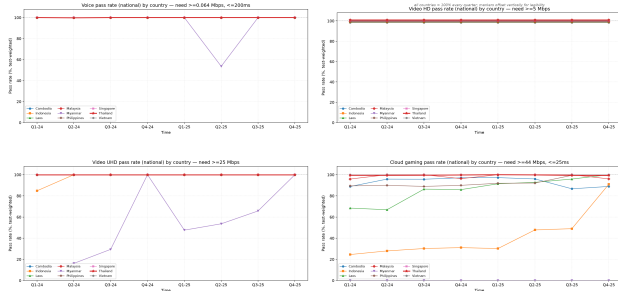
Table 1 lists the bandwidth and latency requirements used as a threshold to evaluate whether the services are good enough for such applications, following Lübben and Misfeld [2].

## 3 DATA ANALYSIS

Across the nine Southeast Asian countries analyzed, Ookla fixed broadband data (2023 Q1 – 2025 Q4) reveals significant regional performance tiers alongside pronounced urban-rural divides when



**Figure 1: Mean download speed of fixed broadband (Ookla)**



**Figure 2: Evaluation results on the application requirements**

comparing capital city performance against national averages. Singapore leads the region overall, with its Central Region reaching 361 Mbps, slightly below its national mean (395 Mbps). In contrast, most other capital regions significantly outperform their nationwide benchmarks: Bangkok Metropolis leads Thailand at 299 Mbps (vs. 277 Mbps national), Kuala Lumpur drives Malaysia to 212 Mbps (vs. 195 Mbps national), Jakarta drives Indonesia to 63 Mbps (vs. 41 Mbps national), the Philippines' NCR reaches 160 Mbps (vs. 132 Mbps national), and Hanoi records 161 Mbps for Vietnam (vs. 160 Mbps national). At the lower end of the spectrum, Cambodia (Phnom Penh at 56 Mbps vs. 53 Mbps national) and Laos (Vientiane Capital at 56 Mbps vs. 51 Mbps national) show modest capital premiums, while Myanmar (Yangon) trails at 22 Mbps, sitting slightly below its national average of 25 Mbps. Overall, these metrics highlight a stark infrastructure gap where the fastest capital (Singapore) reaches speeds 16.4 times greater than the slowest capital (Yangon), with metropolitan centers generally serving as the primary drivers of national connectivity.

For the national-level pass rate for voice and HD video streaming—defined as the test-weighted share of province-quarters achieving a download speed of  $\geq 64$  Kbps and 5Mbps respectively, with voice also requiring latency  $\leq 200$  ms—across nine Southeast Asian countries from Q1 2024 through Q4 2025, nearly every country consistently achieves a 100% pass rate every quarter. The only departures are Thailand's brief dip to 99.8% in Q2 2024 and Myanmar's drop to 53.6% in Q2 2025, both driven by latency rather than bandwidth. This indicates that across all evaluated nations, including lower-tier broadband markets, existing fixed and mobile network infrastructure universally meets the basic 5 Mbps threshold required for smooth HD video playback without performance bottlenecks. However, the national pass rates for Ultra High Definition (UHD) video streaming—defined as the test-weighted share of province-quarters achieving download speeds of  $\geq 25$  Mbps—across the same eight quarters reveal a clear

divergence in performance. Malaysia, Singapore, Thailand, the Philippines, Vietnam, Cambodia, and Laos consistently achieve a 100% pass rate throughout the entire period (clustering together along the top boundary), with Indonesia briefly dipping to 84.8% in Q1 2024 before also settling at 100%. Myanmar consistently trails the region, with its UHD pass rate swinging between 10.5% and 100% over the two-year timeframe. Additionally, the national pass rates for cloud gaming—defined by the test-weighted share of province-quarters clearing both a bandwidth threshold of  $\geq 44$  Mbps and a latency threshold of  $\leq 25$  ms—across nine Southeast Asian nations from Q1 2024 to Q4 2025 show Singapore and Vietnam forming the top tier, both maintaining a near-perfect 100% pass rate throughout, while Thailand and Malaysia fluctuate slightly around 96%–100%. The Philippines and Cambodia occupy a mid-tier position, ranging between 87% and 99%, while Laos fluctuates more widely between 67% and 100%. Indonesia shows a marked recovery, climbing from 24.5% in Q1 2024 to 91.0% by Q4 2025. Myanmar fails to clear the combined threshold in any quarter, holding at 0% throughout: every province-quarter that clears the 44 Mbps bandwidth bar still falls short of the 25 ms latency requirement.

## 4 CONCLUSION

This study provides a multi-year empirical evaluation of network infrastructure performance and application readiness across nine Southeast Asian nations using over 990 million tests from Ookla and M-Lab NDT7 datasets. Our findings reveal that while foundational network requirements—such as voice calls and HD video streaming  $\geq 5$  Mbps—are met with a pass rate at or near 100% across nearly all evaluated countries, significant digital divides emerge under higher-tier demands. Fixed broadband infrastructure exhibits stark regional stratification and severe urban-rural disparities, with the fastest capital reaching speeds up to 16.4 times higher than the slowest capital. While mobile networks provide a more spatially uniform and reliable baseline in developing economies like Myanmar, Laos, and Cambodia, they struggle to sustain bandwidth-intensive, latency-critical workloads. Ultimately, this research underscores that raw speed rankings fail to reflect end-to-end user experiences across diverse application types, highlighting the imperative for targeted fiber deployment, regional infrastructure investments, and transport-layer optimizations to achieve true digital equity across Southeast Asia. **Future work.** We plan to do more analysis in depth as suggested by the reviewers.

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